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Effects of multiple cold-water immersion during pre-season on recovery performance in under-20 male soccer players: a randomised controlled trial

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ABSTRACT

Introduction: This study was designed to evaluate the effects of multiple cold-water immersions performed daily during the preseason period on biochemical, clinical, and neuromuscular aspects of muscle damage of soccer players.

Methods: This two-arm, prospectively registered, randomized controlled trial, blinded to statistician and assessors, was conducted at professional football club facilities. Twenty-three under-20 semi-professionals male soccer players were randomly allocated into cold water immersion group (bathtub with water and ice at 10°C±1°C for 10 minutes) or control group (rest for 10 minutes), every day, after training sessions during a preseason. Primary outcome was change in creatine kinase (CK) concentration, and secondary outcomes were changes in vertical jump performance, strength and perception of recovery at baseline (T0) and after protocol training (T1).

Results: Analysis of Covariance (ANCOVA) showed a statistically significant time-group interactions for CK concentration, with an average reduction of 280.39 U/L (CI95% = -519.14, -41.64; *d* = 0.55) in the cold-water immersion compared to the control group. No differences between groups were observed in any other measures.

Conclusion: Multiple cold-water immersions at 10 ºC for 10 minutes decreases CK concentration but does not change any clinical and neuromuscular markers of muscle damage in soccer players during a 9-day preseason.

Keywords: Cryotherapy, football, performance, fatigue.

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1. Introduction

Soccer imposes a high neuromuscular demand due to changes in direction, acceleration and decelerations, with a predominance of eccentric muscle contraction (Thomas et al., 2017). These demands lead to exercise-induced muscle damage (Barber et al., 2017) evidenced by muscle soreness, extracellular concentration of creatine kinase (CK) and reduced power, strength and sprint ability (Hunkin et al., 2014). High-performance sports work with short rest time exposes athletes to a strenuous training routine, and do not provide players sufficient time to fully recover. The outcome of the training schedule results in reduced performance (Nédélec et al., 2012).

Training loads and recovery period must be managed to ensure soccer players to have adequate performance capacity (Buchheit et al., 2013). Recovery methods have been a topic of interest with the objective of enhancing muscle recovery, particularly during short-term competition with intense training sessions (Nédélec et al., 2013). Cold water immersion (CWI) is a low-cost method that trigger positive physiological effects involving the decrease in cellular metabolism, secondary hypoxia and permeability of the muscle fiber, and increased of hydrostatic pressure and vasoconstriction (Ihsan et al., 2016).

CWI effects have been investigated on laboratory-induced damage (Machado et al., 2017), in real conditions after training and competitions (Dantas et al., 2019; Wilson et al., 2018), however, systematic reviews with meta-analysis have shown conflicting effects on beneficial effects in the muscle pain recovery process (C. Bleakley et al., 2012; Leeder et al., 2012; Machado et al., 2016; Higgins et al., 2017). These conflicting results were observed in single bout CWI studies, however, the use of multiple CWI bouts gained more attention in studies of different sports modalities (De Freitas et al., 2019; Siqueira et al., 2018; Tavares et al., 2019).

In a study examining the effects of multiple CWI sessions in professional volleyball players (5 consecutive days of volleyball training period for 15 min. At 14°C) researchers found no positive effects on muscle damage, physical performance, inflammatory, hormonal and systemic oxidative stress (De Freitas et al., 2019). However, another article applied CWI for three successive days for 20 min. at 10°C after 100 drop jumps in physically active males found a reduction in muscle damage evidenced by reduced muscle soreness and thickness, attenuated CK activity, however this protocol was ineffective on reducing makers of systemic inflammatory and functional performance recovery (Siqueira et al., 2018). Additionally, Tavares *et al*. did not find any deleterious effects nor accelerate performance recovery during 12 consecutive days of intense three-weeks rugby training after multiple CWI (10 min. at 10°C after each day training) (Tavares et al., 2019).

The literature has shown conflicting results in different sports following multiple CWI, however, systematic reviews, including meta-analyses, have addressed gaps in understanding regarding water temperature and immersion time (Machado et al., 2016), as well as immersion depth (Abaïdia et al., 2017; Hohenauer et al., 2020). Protocol used to induce muscle damage, CWI protocol and population training status variability could explain the conflicting findings, limiting the practical applicability of multiple CWI within the sports context. Despite the importance of recovering performance between training and games, no studies have investigated whether multiple CWI promote recover of indirect markers of muscle damage in soccer players in a real training condition. Therefore, the aim of the study was to test the hypothesis that multiple CWI performed daily during a preseason period help to accelerate the recovery of clinical, biochemical and neuromuscular aspects of muscle damage in soccer players.

2. Methods

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2.1. Trial design

We conducted a two-arm, prospectively registered, randomized controlled trial, blinded to statistician and assessors, comprised of under-20 semi-professional male soccer players after approval by Institutional Research Ethics Committee (Number: 851.084) and registered at ClinicalTrials.gov (Identifier: NCT02688972). We conducted this study according to the guidelines of the Declaration of Helsinki and it is reported based on the Consolidated Standards of Reporting Trials (CONSORT) recommendations (Boutron et al., 2017). All participants received oral and written instructions regarding the process and aims of the study and they gave their written informed consent before inclusion.

2.2. Participants

Twenty-three under-20 semi-professional male football athletes volunteered to participate in this study and followed the same physical training, lifestyle and eating habits recommended by the club. They were free of injury and illness at the time of testing and during match participation. Each player was instructed not to perform any additional physical activity, and not to consume alcohol, caffeine, or another nutritional supplementation. Goalkeepers were not included due to the difference in training specificity. Participants were excluded if they showed any sign of musculoskeletal injury during the experiment; missing a training session; or practiced any physical activity during the resting period.

2.3. Procedures

This study started after two days of rest and the experimental protocol was conducted within the preparatory training period, four weeks before the national championship, with a total duration of nine days. The second week of preseason (beginning of annual schedule of training after a vacation period) was chosen to undertake

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the experiment because the team staff believed that players were not yet adapted to training and could suffer the muscle damaging effects of the eccentric-concentric contractions, typical of a soccer training routine.

We adopted an ordinary micro cycle strategy, which is characterized by the regular increase of loads, with 60 and 80% volume and level of intensity. This micro cycle represents a well-known training process for footballers of different levels of performance (Gomes and de Souza, 2008). The training sessions were adapted so that the training load on the days that the initial and final evaluations were standardized.

All players trained once a day (~120 min) for five days during the study (days 1, 2, 3, 7, and 8). They also had a friendly match (day 4) and two rest days on the weekend (days 5 and 6). Experimental design is attached as Supplementary Material. After each training session, we collected the Rating Perceived Exertion (RPE 0-10) of that session. All procedures were carried out at the professional football club facilities during the afternoon (environmental temperature of 26ºC ± 2ºC) with minimal variation in the time schedule in order to avoid fluctuations due to circadian cycle.

2.3.1. Interventions

We randomly allocated the subjects into two groups, control or CWI, through a randomization tool (www.randomization.com). CWI group intervention was carried out in a bathtub with water and ice at 10°C±1°C for 10 minutes. All athletes remained seated with legs in knee extension and the water up to their iliac crest (Goodall and Howatson, 2008). When necessary, water at room temperature, or ice, was added to the bathtub to take the temperature back to 10°C in case it decreased or increased, respectively.

Athletes allocated into the control group remained seated on the floor with legs in knee extension during 10 minutes, same time and position of athletes in the CWI group.

2.3.2. Outcomes measures

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Assessments at baseline (T0) and one day after finish of training protocol (T1) were performed by a blinded researcher. In the first day of the experiment, the soccer players were evaluated with anthropometric data, food consumption, and sleep quality surveys using the Pittsburgh Sleep Quality Index (Bertolazi et al., 2011). After that, players were evaluated regarding dependent variables of study: biochemical indicators of muscle damage (CK), functional performance (jumping height and isometric muscle strength of the knee flexors and extensors) and quality of the recovery process.

Peripheral blood samples were collected in anti-coagulant tubes (8 ml) to analyse blood CK levels. CK samples were centrifuged for 10 minutes at 2500 rpm to obtain the plasma in tubes containing K2EDTA. Plasma was stored at −20°C until the time of analysis. CK (CK NAC Liquiform, Ref.:117, Labtest, Minas Gerais, Brazil) was measured using colorimetric methods (UV kinetic and Pyruvate-Lactate reactions, respectively) on a biochemical analyser Labmax Plenno (Labtest, Minas Gerais, Brazil).

For Vertical Jump (VJ) maximum height analysis, a contact mat (1000 x 600 x 8 mm; Jump System Pro) was used. The participants were positioned in the mat with hands placed on the iliac crest. Then, they flexed the knee and performed a maximum vertical jump initializing and finishing the movement with both feet on the contact mat (Kenny et al., 2012). The athletes were instructed to perform 3 vertical jumps with 10 seconds intervals between repetitions and the highest score was used for analysis.

To measure the maximum voluntary isometric force (MVIF), the subjects were positioned in an adjustable chair with their hips at 90º and with their trunk and contralateral lower limb stabilized with a belt. The distal shin pad of the dynamometer was positioned 3 cm above the lateral malleolus and fixed by a strap. To analyse knee extensor or flexor muscles, an angle of 60º and 30º, respectively, were adopted where 0º were considered the complete extension. Then, we performed three maximum voluntary

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isometric contractions; each contraction lasted 5 seconds with intervals of one minute and the highest score was used for further analysis.

We applied the Total Quality Recovery perceived (TQRp) scale to measure the recovery process. This data was collected every day athletes were exposed to interventions. Each day, before bedtime, participants were asked to rate their recovery as an overall psychophysiological rating for the previous 24 hours, including the previous night's sleep (Kenttä and Hassmén, 1998). The scores of this scale varied from 6 to 20 where the higher the score indicating the greater recovery perceived of the athletes.

2.4. Sample size

A priori sample calculation was performed using *G* Power software* (version 3.1.3; University of Trier, Trier, Germany) (Beck, 2013). We adopted an $\alpha = 0.05$, power $(1-\beta) = 0.90$, correlation coefficient = 0.5 and effect size =0.28 (De Freitas et al., 2019). Based on these values it was estimated that a total of 20 subjects, 10 per group, would be an adequate sample size, but 23 players were included for analyses.

2.5. Randomization, allocation and blinding

Offsite randomization was performed and allocation concealment was ensured using sealed opaque envelopes inaccessible to blinded assessors. After signing the informed consent form, eligible individuals were randomly allocated to the intervention or control group by another independent researcher after baseline assessment. All analyses were performed by an independent researcher blinded to group allocation, and individuals were asked not to reveal their allocation status during the study period. In addition, at any moment of study participants receive information about which technique could be more effective for the muscle recovery process.

2.6. Statistical analysis

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A statistician blinded to the group allocation conducted all analyses before decoding the groups. Statistical analysis was performed using the SPSS Version 24.0 (SPSS Inc, Chicago, IL). Shapiro-Wilk and Levene tests showed a normal distribution of data and homogeneity of variances, respectively. Then, Mauchly's test was used to measure sphericity, and in case of sphericity violation, Greenhouse-Geisser test was conducted to perform corrections. An Analysis of Covariance (ANCOVA) was used to analyse time-group interactions of all variables, using T0 values of each variable as covariable. Cohen's *f* was used to assess the effect size of interaction (small at 0.10, medium at 0.25 and large at 0.40). Main results were expressed in mean difference and confidence interval (CI95%), at a 5% significance level and 95% confidence interval. Cohen's *d* coefficient was used to measure the paired effect size of the differences between groups. An effect size (ES) greater than 0.8 was considered large; between 0.5 and 0.7, moderate; from 0.4 to 0.2, small; and 0.1 or < 0, has no effect.

3. Results

27 participants were available to participate in the study, but only 23 met all eligibility criteria and were randomised (CONSORT Flowchart attacked as Supplementary Material 2). Groups were similar at baseline regarding anthropometric variables, food consumption and sleep quality between groups (Table 1).

INSERT TABLE 1

During the 6 days of training, the rating perception exertion (RPE) showed an average of 4.84 (CI95% 4.52, 5.15) for CWI group and 4.67 (CI95% 4.33, 5.01) for the control group (mean difference = 0.16; CI95% = -0.30, 0.63) (Figure 2). The RPE's means of the groups showed similar behaviour during most training days. On day 3, we found an average reduction in groups of 2.5 units (CI95% = 1.47, 3.53; *f* = 0.77) compared to day 2, and a decrease of 2.6 units (CI95% = 1.26, 3.99; *f* = 0.77) compared to day 1.

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On day 5, there was an increase of 2.33 units (CI95% = -4.14, -0.52; d = 1.77) compared to day 3, only for the control group.

INSERT FIGURE 1

We found a reduction os CK levels of 280.39 U/L (CI95% = -519.14, -41.64; *d* = 0.55) in the extravasation of CK in the CWI group compared to control in T1 assessment (Table 2). In addition, a statistical interaction time-group was observed for CK concentration ($F_{1,23}$ = 5.902; p = 0.023; *f* = 0.50) (Table 2) after multiple CWI interventions during 10 days of training.

INSERT TABLE 2

We did not find time-group interaction for vertical jump (Height: $F_{1,23}$ = 0.141; p = 0.711; *f* = 0.07; Power: $F_{1,23}$ = 0.304; p = 0.587; *f* = 0.11) (Table 2). There was an average reduction of 0.41 cm (CI95% = -2.72, 1.88; *d* = 0.31) in height of vertical jump in the CWI group compared to control in T1 assessment. CWI group showed a reduction of 0.50 watts (CI95% = -2.41, 1.39; *d* = 0.29) in comparison to the control group on power of vertical jump (Table 2).

We did not observe time-group interaction for MVIF (Extensor: $F_{1,23}$ = 0.329; p = 0.572; *f* = 0.11; Flexor: $F_{1,23}$ = 0.023; p = 0.881; *f* = 0.03) (Table 2). In T1 assessment, there was a small improvement of 2.38 (CI95% = -6.20, 10.96; *d* =0.34) in extensor MVIF in the CWI group compared to control (Table 2). Moreover, we observed a small difference of 0.31 (CI95% = -3.91, 4.53; *d* = 0.36) between groups in T1 assessment, which CWIG showed best values (Table 2).

Regarding TQRp, we did not found statistical time-group interaction ($F_{(6.11)}$ = 0.435, p = 0.854) (Figure 2). Moreover, the groups showed different perception of recovery at day 3 (mean difference = -1.73; CI95% = -3.27, -0.29; *d* = 0.97), day 5 (mean difference = -2.42; CI95% = -3.55, -1.28; *d* = 1.74) and day 6 (mean difference = -2.61;

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CI95% = -4.31, -0.91; *d* = 1.25) in which the control group had the worst subjective perception index of recovery (Figure 2).

INSERT FIGURE 2

4. Discussion

While the acute effects of CWI have been previously investigated in soccer players, it is the first study to analyse biochemical, functional and neuromuscular performance during a pre-season of multiple exposure to CWI. In addition, our study exhibits a well-structured methodological design characterised by a randomized, prospectively registered, masking of the statistician and evaluators, and allocation concealment. The main finding from the current study is that the multiple exposure to cold water immersion over a pre-season phase in soccer players demonstrated an effect just on biochemical markers of muscle damage. Despite multiple CWI not showing an effect on the recovery of functional and neuromuscular parameters, this method also exhibited no detrimental effect to performance.

Soccer is a sport modality characterized by an intense neuromuscular demand such concentric and eccentric muscle activation and a high musculoskeletal effort (Thomas et al., 2017), however the first 2 weeks of pre-season was characterized by low-to-moderate muscle damage due average RPE during period of study. One of the possibilities of the non-effect of the intervention in all indirect muscle damage markers may involve the magnitude of muscle damage produced by the pre-season once physiological stress include athletes' acclimation to the intensity and duration and of the exercise (Leeder et al., 2012).

Regarding biochemical analysis, CWI group showed a decrease in CK serum levels after multiple sessions, in contrast with previous studies, which assessed the effect of multiple CWI after drop jumps (Goodall and Howatson, 2008). Our results were similar

to Ascensão *et al.* (Ascensão et al., 2011) using the same CWI protocol (10 ºC for 10 minutes) in soccer players but developed after one soccer match performing a single session of CWI, and to Siqueira *et al*. (Siqueira et al., 2018), after carried out 100 drop jumps followed by multiple CWI. This result suggests the possible change in blood flow by reduction of metabolic activity and secondary hypoxia through vasoconstriction caused by CWI (Ihsan et al., 2016) may explain a reduction of 280.39 U/L at the final of pre-season for the CWIg. It is also believed that CWI decreases lymphatic permeability resulting in reduction of diffusion attenuating inflammatory processes (Wilcock et al., 2006).

Multiple CWI decreased biochemical muscular damage without reducing the athlete's performance and adaptation process. Our result is corroborated with previous work, which Eston & Peters (Eston and Peters, 1999; Goodall and Howatson, 2008; Tavares et al., 2019; De Freitas et al., 2019) did not find effect of multiple CWI on strength, and functional performance.

The literature suggests that by studying subjects at the pre-season phase can interfere with the performance based on recruitment, frequency and synchronism of motor unit firing (Lavin et al., 2022), but probably due moderate mean RPE, the performance was maintained and some variables was improved. Chronic use of CWI could maintain these neural aspects with increasing exercise tolerance (Algafly and George, 2007) and restricting the entry of inflammatory cells into the muscle by decreasing vascular perfusion and reducing local metabolism (Roberts et al., 2014). However, CWI does not seem to interfere in the metabolic and neuromuscular aspects that could decrease strength and functional performance after muscle damage once tissue temperature reduction in humans does not influence cellular metabolic activity (C. M. Bleakley et al., 2012). In

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addition, CWI could impair the sports performance since it can alter the recruitment of motor units and the capacity of generating strength (Herrera et al., 2010).

Regarding perception of recovery, CWIg showed better levels on days 3, 5 and 6, in contrast with other studies that found no difference in TQRp scores comparing swim and passive recovery after high intensity interval exercise (Lum et al., 2010). Our results corroborated with the results of Suzuki *et al.* (Suzuki et al., 2004) who reported improved psychological ratings, however used aquatic exercise in rugby athletes and assessed by Profile of Mood State questionnaire, and with Brophy-Williams *et al*. (Brophy-Williams et al., 2011) who found better TQRp scores for CWIg group compared with control group (rest). Both studies (Brophy-Williams et al., 2011; Suzuki et al., 2004) used just one CWI session and levels of improvement in perceived recovery may be related with reduction of nerve conduction velocity (Tashjian et al., 2009).

4.1. Limitations of the study

The main limitations of the present study were: a) impossibility of masking the therapist; b) we did not collect the level of delayed onset muscle soreness and the participant's expectation about effects of CWI as it can influence some of the outcomes. Another limitation was the non-inclusion of a water immersion (Dantas et al., 2019) or a placebo group (Broatch et al., 2014) as studies have shown to be as effective as CWI in attenuating CK extravasation, maintaining functional performance and improving subjective measures.

5. Conclusion

Multiple CWI at 10ºC for 10 minutes is effective for the biochemical aspect of muscle damage. In addition, multiple CWI does not accelerate functional and neuromuscular recovery markers, and also did not impair performance.

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5.1. Clinical relevance

- This is a high methodological quality study because it is randomised, prospectively registered, with masking of the statistician and evaluators, and with allocation concealment. This experimental design helped us to measure with confidence the effects of multiple exposure to CWI on biochemical, functional and neuromuscular performance during semi-professional soccer pre-season;
- Multiple CWI can be used during pre-season aiming attenuate the biochemical characteristics of muscle damage;
- We must be careful in extrapolating the results to other aspects of muscle damage; multiple CWI showed no effects to improve functional and neuromuscular recovery, but also did not interfere negatively.

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CRediT authorship contribution statement:

SMP was involved in drafting this paper and participated in the conception, study design, assessments, data interpretation, writing and submission of the manuscript. SMP, GAFD, GMB and PMSD contributed to the study design, data interpretation and writing the manuscript. SMP, GAFD, LDRS, GST, GMB and PMSD took part in management, analysis, data interpretation and writing the manuscript. All authors read and approved the final manuscript. SMP takes the responsibility for the integrity of the work as a whole.

Conflict of interest

None.

Patient consent for publication

Obtained.

Ethics approval

This study was approved by the Institutional Ethics Committee of the Federal University of Rio Grande do Norte.

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Tables

Table 1 - Baseline characteristics of the participants.

	CWIg	**Cg**
Age (years)	17.9 ± 0.7	18 ± 0.8
Body Mass (kg)	73.2 ± 9.5	68.0 ± 4.6
Height (m)	1.8 ± 0.1	1.7 ± 0.1
Body Mass Index (kg/m^2)	22.3 ± 1.3	22.1 ± 0.9
Energy (kcal)	3164.4 ± 1095.2	2693.2 ± 587.9
Carbohydrates (kcal)	1705.4 ± 608.9	1513.3 ± 276.5
Proteins (kcal)	494.8 ± 115.3	443.0 ± 89.9
Lipids (kcal)	949.5 ± 489.1	729.1 ± 321.7
Sleep quality (%)	75%	69%

Values are mean ± standard deviation. **CWIg:** Cold-Water Immersion group; **Cg:** Control group

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Table 2 - Muscle damage indirect markers during period of study.

	T0	T1	Mean Difference at T1 (CI 95%; Cohen's *d*)	Interaction effect
CK (U/L**)**				
CWI	254.67 ± 94.03	608.98 ± 77.64	280.39 (-519.14, -41.64)	$F_{1,23}$ = 5.902; p = 0.023;
Control	254.67 ± 94.03	889.37 ± 83.98	*d* = 0.55	*f* = 0.50
Heigth VJ (cm**)**				
CWI	35.73 ± 5.47	36.66 ± 0.75	0.41 (-2.72, 1.88)	$F_{1,23}$ = 0.141; p = 0.711;
Control	35.73 ± 5.47	37.08 ± 0.81	*d* = 0.31	*f* = 0.07
Power VJ (Watts**)**				
CWI	46.87 ± 4.91	47.54 ± 0.62	0.50 (-2.41, 1.39)	$F_{1,23}$ = 0.304; p = 0.587;
Control	46.87 ± 4.91	48.05 ± 0.67	*d* = 0.29	*f* = 0.11
Extensor MVIF (kgf**)**				
CWI	57.05 ± 12.46	65.96 ± 2.71	2.38 (-6.20, 10.96)	$F_{1,23}$ = 0.329; p = 0.572;
Control	57.05 ± 12.46	63.58 ± 2.94	*d* =0.34	*f* = 0.11
Flexor MVIF (kgf**)**				
CWI	29.50 ± 7.21	29.45 ± 1.36	0.31 (-3.91, 4.53)	$F_{1,23}$ = 0.023; p = 0.881;
Control	29.50 ± 7.21	29.14 ± 1.47	*d* = 0.36	*f* = 0.03

Legend: Analysis of Covariance (ANCOVA) using baseline assessment (T0) as covariable. **T0:** baseline assessment; **T1:** assessment at one day after finish of training protocol; **CK:** creatine kinase; **MVIF:** maximum voluntary isometric force.

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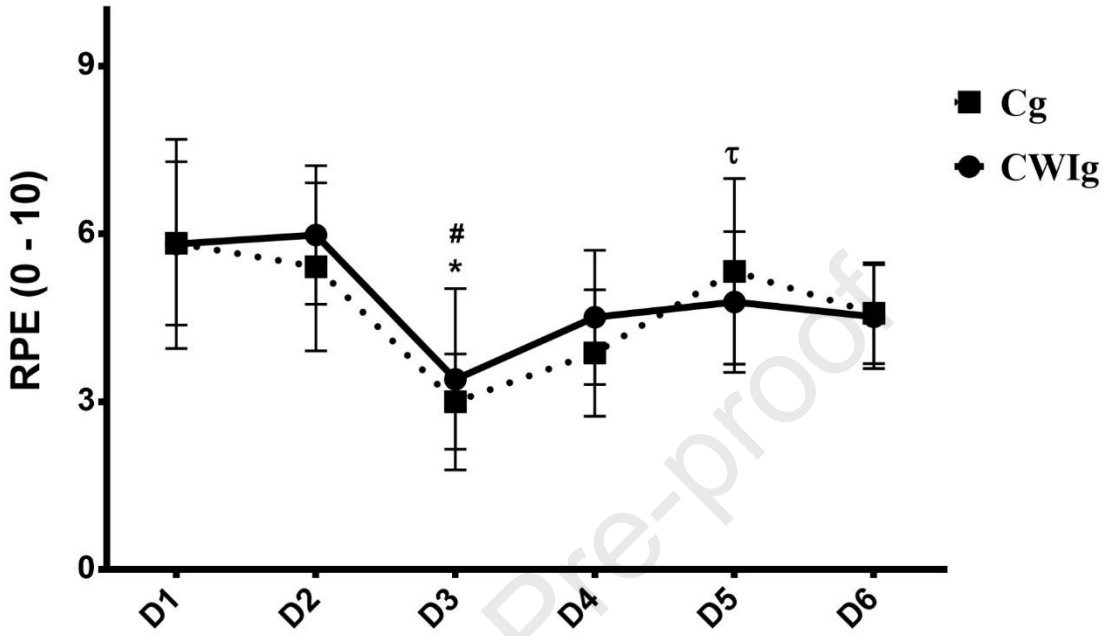
Figures captions

Figure 1



Legend: CWIg: Cold-water immersion group; **CG:** Control group; ***:** $p<0.05$ compared to day 2; **#:** $p<0.05$ compared to day 1; **τ:** $p<0.05$ compared to day 3 (just control group).

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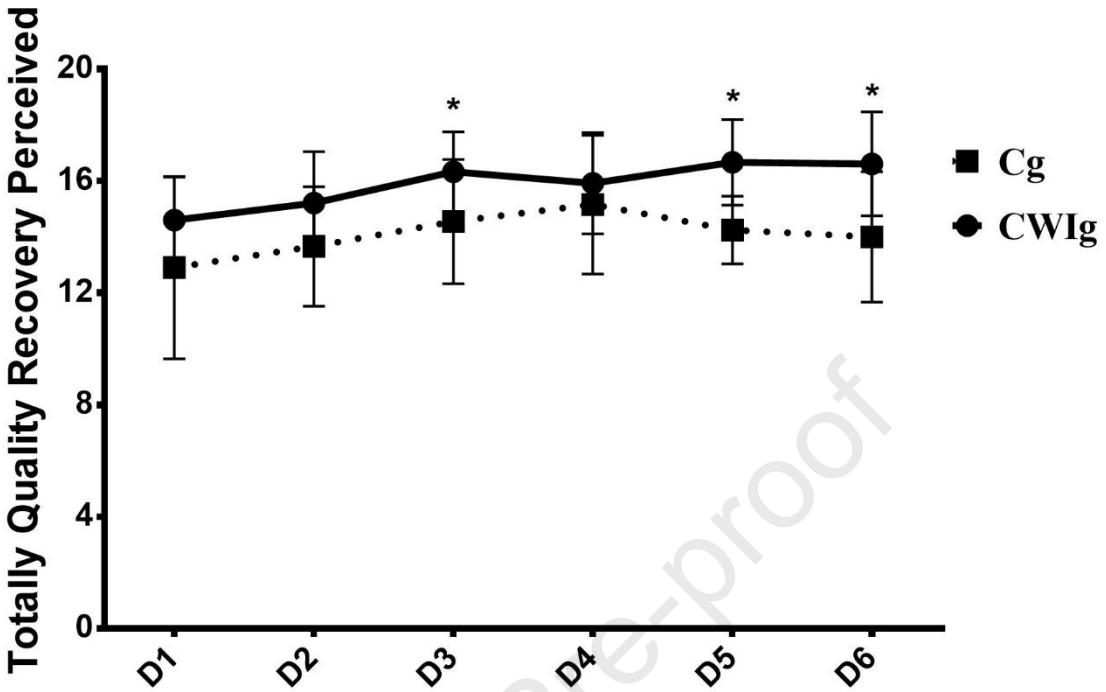
Figure 2



Legend: CWIg: Cold-water immersion group; **CG:** Control group; ***:** $p<0.05$ between group difference.

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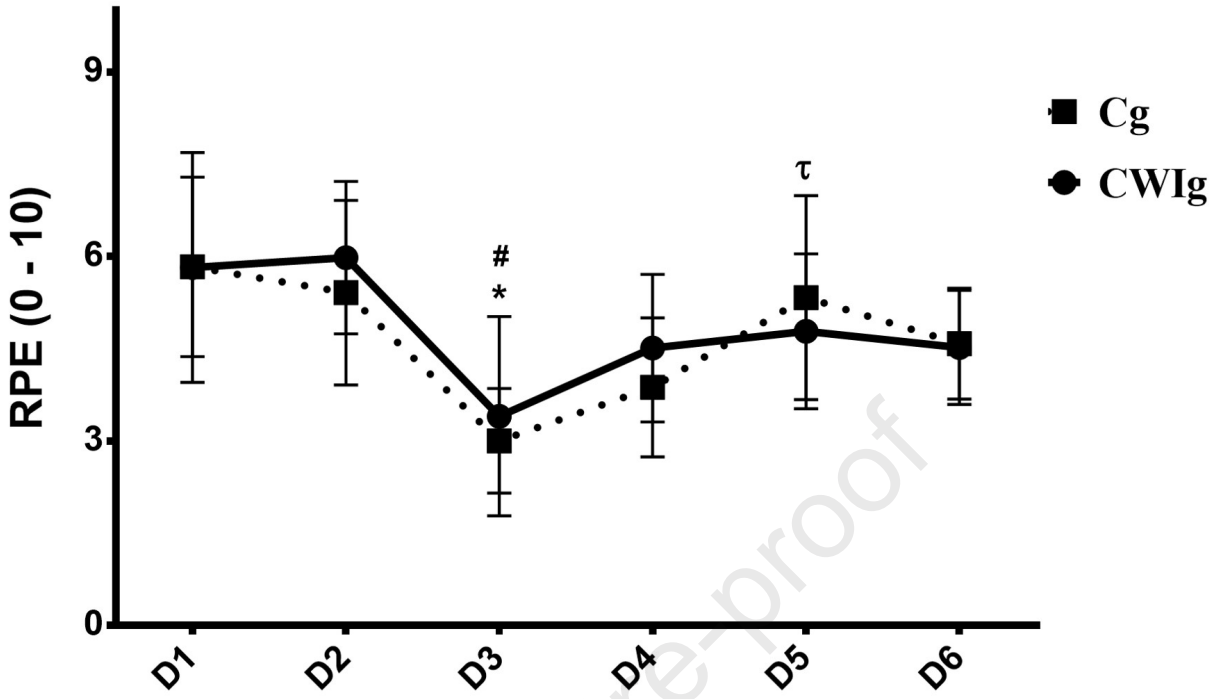

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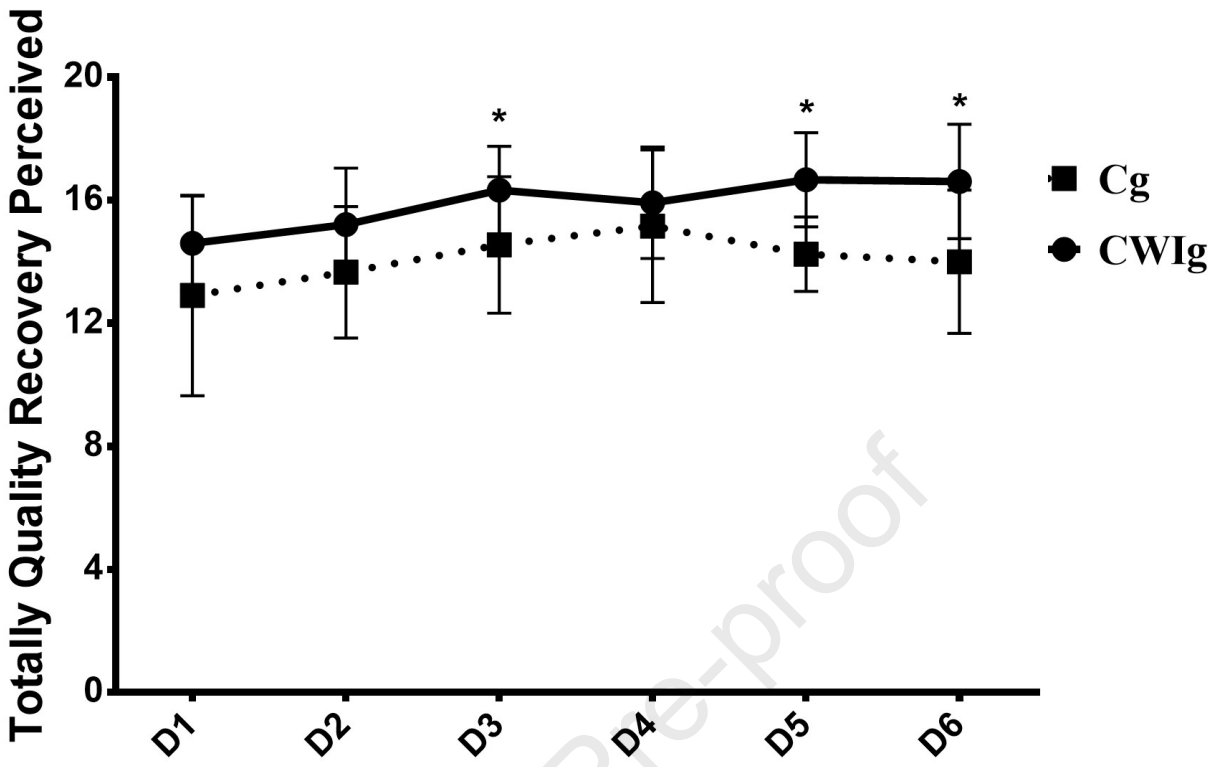

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